INCH-POUND

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SUPERSEDING
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24 March 1976

MILITARY SPECIFICATION

KETTLES, STEAM JACKETED; TWIN; 5-GALLON, TILTING, CABINET MOUNTED; NATURAL GAS HEATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two 5-gallon capacity, tilting, steam jacketed kettles, cabinet base, and a self-contained, gas heated, steam generator boiler.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-T-60 - Tape: Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN GAS ASSOCIATION, INC. (AGA)

Directory of Certified Appliances and Accessories

(Copies should be obtained from the American Gas Association, Inc., 8501 East Pleasant Valley Road, Cleveland, OH 44129.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI - Z83.15 Kettles, Steam Cookers, and Steam Generators

(Copies should be obtained from the American National Standards Institute, 1430 Broadway, New York, NY 10018, or the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44129.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code

Section IV - Rules for Construction of Heating Boilers

Section VIII - Unfired Pressure Vessel Code

(Copies should be obtained from the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 240 - Stainless and Heat-Resisting Chromium and Chromium Steel Plate, Sheet and Strip for Fusion-Welded Unfired Pressure Vessels

D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Current Listing of Food Service Equipment

(Copies should be obtained from the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 197 - Commercial Electric Cooking Appliances

(Copies should be obtained from the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The tilting kettle assembly delivered under this document shall be the manufacturer's commercial product, except for any changes necessary to comply with requirements specified herein. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.2 Codes and standards. The tilting kettle assembly shall comply with the applicable requirements of UL 197, ANSI Z83.15; the ASME, Sections IV and VIII; and NSF Standard No. 4, as applicable (see 4.3).

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial tilting kettle assembly, provided the finished item complies with all provisions of this document. (It is encouraged that recycled material be used when practical as long as it meets the requirements of this document. (see 6.3)

3.3.1 Stainless steel.

3.3.1.1 Plate, sheet, and strip. Stainless steel plate, sheet, and strip shall be 304 of ASTM A 240.

3.4 Design and construction. The tilting kettle assembly shall consist of two 5-gallon capacity stainless steel steam jacketed tilting kettles with trunnion(s) mounted on an enclosed cabinet base. The cabinet base shall be provided with doors and shall house a self-contained ASME approved steam generator boiler, piping, gage, and controls necessary for safe and efficient operation. The steam generating boiler shall be gas heated. The completely assembled unit (i.e. cabinet base with steam generating boiler and tilting kettles) shall be ready for connection to drain, water, gas, and electrical supplies. The tilting kettles shall be designed to cook, simmer, or keep food products warm in a temperature range of 150°F to not less than 240°F by means of steam generated in the steam generating boiler. Dimensions and connection sizes for the tilting kettle assembly and steam generating boiler, shall conform to table I. All kettles, trunnion mounting(s), steam generator boiler, piping, joints, and fittings shall be assembled to provide leakproof components and joints (see 4.4.1).

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TABLE I. Dimensional requirements

Requirement	Dimension and connection size (inches)
Overall outside diameter at kettle rim	12 to 15
Height from bottom of kettle mount to kettle rim	15 to 20
Clearance between kettle spout (in full tilt position) and top of stand, minimum	4-1/2
Size of hot and cold water inlet (nominal tubing size)	3/8
Size of gas connection	3/4 or 1 Iron Pipe Size (IPS)
Size of drain connection	1-1/2 or 2 IPS
Size of safety valve connection	3/4 IPS (nominal)
Clearance between top of kettle and swing spout outlet	1-1/2 to 3
Overall dimensions kettles and cabinet base	Width - 35 to 37 Depth - 33 to 37 Height - 53 to 62

3.4.1 Tilting kettles. Tilting kettles shall be designed in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code and shall have a design pressure not less than 30 pounds per square inch gage (psig). The tilting kettles shall be thermostatically operated at any steam jacket pressure up to 15 psig. The tilting kettles shall consist of a kettle body, steam jacket, tilting handle, trunnion mounting(s), safety valve, and plumbing.

3.4.1.1 Kettle body. The body shall be cylindrical with a hemispherical or sloping bottom and shall be fabricated of stainless steel as specified in 3.3.1.1. Material thickness of that section of the body forming the inner wall of the steam jacket shall conform to the requirements of ASME for the rated working pressure of the kettle. The body shall be seamless drawn or spun from one piece of material. The top edge shall be formed into a minimum 5/16 inch, half round, open radius, a fully closed sealed bead, or a double hem. The kettle shall have a pouring lip either formed or welded to the top edge to facilitate pouring when the kettle is tilted.

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3.4.1.2 Steam jacket. The steam jacket shall be stainless steel. Material thickness shall conform to the requirements of ASME for the rated working pressure of the kettle. The jacket shall be formed or spun from one piece or shall be fabricated by welding preformed segments. The top of the jacket shall be welded to the kettle body. A means for complete drainage of the water jacket shall be provided.

3.4.1.3 Tilting handle. The tilting handle shall consist of a stainless steel rod and a heat resistant plastic knob. The handle shall be positioned on the outboard side of each kettle when facing the kettle front, and the rod shall be inserted into the trunnion shaft or into a lug welded to the kettle body above the steam-water jacket. The knob shall be a push-on type or be threaded to receive the threaded end of the rod.

3.4.1.4 Kettle mountings. Kettles shall be provided with either one or two trunnions welded to the kettle body. When one trunnion is provided, the kettle shall be mounted in a combination kettle support and tilting trunnion. The combination support shall be fabricated of stainless steel or cast aluminum finished as specified in 3.8. When two trunnions are provided, the trunnion shafts shall be welded to the body above the steam-water jacket and the trunnion bushings shall be mounted on two stainless steel leg supports. The kettle provided with a combination support shall be equipped with a stainless steel base plate. Both types of kettle mountings shall have provisions for bolting the kettle assembly to the cabinet specified in 3.4.2. The trunnion(s) shall allow 95 degrees of tilt from the normal position and shall allow the kettle to return to the upright position when released. The trunnions and supports shall be capable of supporting a load equivalent to 1-1/2 times the weight of the kettle's water capacity, without deformation when tested as specified in 4.4.4.

3.4.1.5 Cover. Each kettle shall be provided with a one-piece removable cover, formed from minimum 0.056 inch thick stainless steel specified in 3.3.1.1. The cover shall be flanged, dished, or suitably reinforced to prevent distortion as a result of normal use and shall be provided with a knob or loopbar type handle. The top of the kettle body shall fit the flange on the cover.

3.4.2 Cabinet base. The cabinet base shall be of the totally enclosed type enclosing a steam generator boiler and shall consist of a stainless steel base, marine edged cabinet top, single piece lift-off door or two hinged doors, and four adjustable legs. The cabinent top shall be provided with provisions for mounting each tilting kettle, trunnion(s), single swing spout faucet with shut-offs (hot and cold water), and steam valves. The cabinet top shall include, for each tilting kettle, a sink with drain, removable grate, and removable drip tray. The cabinet base top contain a galvanized or standard, welded, angle-iron frame enclosed by an outer skin of not less than 0.0625 inch (16 gage) thick stainless steel specified in 3.3.1.1. The single piece lift-off door or hinged cabinet doors shall be equipped with a handle

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and a magnetic latch assembly, including a ferrous striker plate, mounted, for magnetic attraction to hold the door closed. The cabinet doors shall provide access to the steam generator boiler, controls, valves, and plumbing. The cabinet doors shall be fabricated from not less than 0.0625 inch (16 gage) stainless steel specified in 3.3.1.1. The cabinet top shall be fabricated from not less than 0.0500 inch (18 gage) thick stainless steel specified in 3.3.1.1.

3.4.2.1 Steam generator boiler. The steam generator boiler shall be designed in accordance with Section IV of the ASME Boiler and Pressure Vessel Code for operation at pressures not exceeding 15 psig.

3.4.2.1.1 Safety valve. Each kettle shall be equipped with a pressure-type, brass, safety valve which conforms to the requirements of ASME Boiler and Pressure Vessell Code for the rated working pressure of the kettle. The valve shall be set at the factory in accordance with the ASME Code Section VIII for the maximum working pressure of the kettle. The valve shall be furnished with a reversible lifting level and shall be capable of being rotated 360 degrees without disturbing the valve settings when tested as specified in 4.4.1. The valve shall be provided with a bottom inlet and side discharge outlet and shall be connected to the steam jacket or the steam supply piping within the cabinet.

3.4.2.2 Legs. The cabinet shall be mounted on four, 6-inch high stainless steel legs with adjustable bullet feet. Foot adjustment shall be not less than 1/2 inch. Legs may be either removable or permanently attached by welding to the frame.

3.4.2.3 Controls. Each kettle shall have a manually adjustable steam inlet control valve to control the steam supply from the boiler to the kettle. The control valve(s) shall operate without binding and shall show no evidence of steam leakage when tested as specified in 4.4.1.

3.4.2.3.1 Steam inlet control valve. The steam inlet control valve shall not leak steam or hot water. There shall be no evidence of binding or sticking of the valve during operation (see 4.4.1).

3.4.2.4 Pipe. Pipe shall be IPS with American Standard National Taper Pipe threaded (NPT) ends. Pipe for water lines shall be seamless brass, standard weight.

3.4.2.5 Pipe fittings. Pipe fittings used with brass pipe shall be bronze, and of the type and weight normally furnished by the manufacturer on his standard product. Pipe fittings welded to the steam-water jacket shall be stainless steel when the jacket is stainless steel.

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3.4.2.6 Globe valve(s). Globe valve(s) shall be rated for hot or cold water pressures up to 125 psig. Valve(s) shall be of composition bronze and shall have a removable disc-holder with removable discs suitable for water temperatures up to 212°F. Valve bodies shall have NPT threaded ends. The working pressure shall be cast on the side of the valve body as acceptable evidence of compliance with pressure requirements.

3.4.2.7 Pressure gage. The pressure gage shall be of the dial-type and shall be graduated in pounds from 0 to 30 psig minimum and numbered at 15 pounds or less. Dial face shall be not less than 2 inches in diameter. The pressure gage shall conform to the requirements of ASME Boiler and Pressure Code, for the rated working pressure of the kettle; this shall not vary more than plus or minus 1 psi for the actual steam generator boiler pressure (see 4.4.1). The pressure gage shall be located so as to be visible to the operator.

3.4.2.8 Swing-spout. A swing-spout water supply connection shall be provided. The spout shall be a combination swing-spout and faucet or a swing-spout with globe valves. The swing-spout shall be the manufacturer's standard design and shall be chrome-nickel-plated brass or stainless steel. The swing-spout shall be mounted on the cabinet between each of the 5-gallon tilting kettles. When a faucet is not an integral part of the swing-spout, globe valves (see 3.4.2.6) of the same size as the water connections (hot and cold) shall be installed in the plumbing directly below the swing-spout. The spout outlet shall be provided with an aerator.

3.4.2.9 Automatic drain unit. When specified (see 6.2) the gas steam generator boiler shall be provided with an automatic drain unit, automatic drain units shall be automatically filled and brought to pressure whenever heat and water controls are turned on and while under pressure and shall automatically drain whenever heat and water switches are turned off when tested as specified in 4.4.1. A cold water condenser shall be provided. The cold water condenser shall reduce the temperature of water entering the drain line.

3.4.3 Gas characteristics. The gas steam generator boiler shall operate on natural gas. Burner gas valves and pilot gas valves shall be of the universal or fixed orifice type. Valves of the fixed orifice type shall be provided with interchangeable orifice hoods or bushings drilled for natural gas. The valve hoods for natural gas shall be unpainted brass. The boiler shall have a minimum input of 170,000 British thermal units (Btu) (see 4.4.1).

3.4.3.1 Pressure regulator. A pressure regulator shall be furnished for use with natural gas fuel and the incoming water column gas pressure shall be between 3-1/2 and 7 inches for natural gas.

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3.4.3.2 Automatic ignition. Burners shall be equipped with a means for automatic ignition of gas. A 100 percent shut-off device is required for all automatic ignition of gas and for all burners. The safety shut-off device shall automatically shut off the gas to all burners including pilot burners in event that means of ignition become inoperative (see 4.4.1).

3.4.4 Electrical characteristics. Unless otherwise specified (see 6.2), the electrical components for the tilting kettle assembly shall be suitable for operation on a nominal 120 volt 60 hertz (Hz) system.

3.5 Repair and maintenance. Provisions shall be made to allow access to inspect, service, replace, and adjust any component of the tilting kettles and steam generator boiler. Adjustment, repair, or replacement of any component shall not require disassembly or removal of additional component(s).

3.6 Marking.

3.6.1 Data-name plates. The kettles shall be furnished with a data-name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data-name plates shall be made of minimum 20 gage corrosion-resisting metal and attached to the kettles by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information, which shall be stamped, engraved, or applied by photosensitive means:

National stock number
Procurement instrument identification number
Specification data
Manufacturer's name, address, and telephone number
Supplier's name, address and telephone number (List if different from manufacturer)
Manufacturer's model number
Government approved manual number (see 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the item.

3.6.2 User's instruction plate. Operating instructions shall be on an aluminum, brass, or stainless steel plate permanently affixed to the item. Marking shall be stamped, embossed, engraved, or applied by photosensitive means. The plate thickness shall be not less than 0.012 inch. The plate shall be of sufficient overall size to contain, in legible characters, the

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information listed below. The words "STEAM-JACKETED KETTLE" shall be not less than 3/16 inch high and the words "USER'S INSTRUCTIONS" shall be not less than 1/8 inch high. All other characters shall be not less than 3/32 inch high. The plate shall be mounted on the inner door of the cabinet.

STEAM-JACKETED KETTLE

USER'S INSTRUCTIONS

1. Caution: Steam supply shall not exceed 15 psig.

2. Do not tamper with or obstruct safety valve.

3. Do not clean with abrasives.

4. Wash kettle before and after each use with hot soapy water and rinse thoroughly with clean hot water.

5. Do not turn steam on unless water or food is in the kettle.

6. Do not put water in hot dry kettle.

7. When heating a cold kettle, turn the steam on gradually, thus allowing the kettle to become warm before full pressure is applied. After kettle becomes warm, and before applying full pressure, open the safety valve momentarily to release any air trapped within the jacket.

3.7 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, or other foreign matter. Welds shall be sound and free from porosity, cracks or incomplete fusion. Exposed welds shall be free from burrs or sharp projections. Welding shall be made with welding rods having the same chemical composition as the welded material. The finished weld shall be smooth and uniform.

3.8 Finish. The tilting kettle assembly finish shall conform to NSF Standard No. 4 and as follows: Exposed piping and fittings, hinges, and latches when not stainless steel or aluminum shall be chrome-plated in accordance with the manufacturer's standard practice. Standard welded angle iron frame when provided, shall have a rust-proof painted finish.

3.9 Workmanship. Components shall be free of all slivers, burrs, and rough or sharp edges. Bolt holes and rivet holes shall be accurately punched or drilled. Washers or lockwashers shall be provided in accordance with standard commercial practice, and all bolts, nuts, and screws shall be drawn tight. Rivets shall be tightly driven and fully peened.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of tilting kettle assemblies. The sample unit shall be one tilting kettle assembly. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Design, construction, and workmanship (general)	Not finished in accordance with NSF Standard No. 4	101	
	Specified components not chrome-plated		201
	Fractured, dented, or bowed component	102	
	Rough or sharp edges, burrs, or slivers	103	
	Component not readily accessible for servicing where required		202
	Standard welded angle iron finish not rustproof	104	
Casting (where required)	Cracked, porous, blowholes, or other casting imperfections	105	
Welding (where required)	Incomplete, burn holes, cracked, fractured, or not fused	106	
	Slag inclusion, slight undercut, scale or flux deposits not removed		203
	Weld not smooth or nonuniform cross-section	107	
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, broken, stripped, or fractured	108	
	Loose		204
	Not equipped with washers or lockwashers		205
Rivets	Loose; head not in full contact with member		206
Kettle	Body bottom not hemispherical or sloping	109	
	Not equipped with pouring lip	110	
Jacket	Not equipped with means for draining	111	
Tilting handle	Not positioned on outboard side of each kettle	112	
	Not located above steam-water jacket	113	
	Not equipped with heat resistant knob	114	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Mount	One or two trunnions not properly welded to kettle body	115	
	Not bolted to cabinet base	116	
	Will not allow 95 degrees of kettle movement	117	
Cover	Not one piece construction	118	
	Not removable type	119	
	Not as specified (i.e., flanged)	120	
	Missing	121	
	Handle not knob or loop-bar type	122	
	Handle loose		207
Cabinet base	Adjustable legs missing or not as specified	123	
	Frame not angle-iron	124	
	Top not provided with sink, drains, removable grates, removable drip-trays, trunnion mounting(s), swing-spout, or faucet	125	
	Top not marine edged type	126	
Generator	Not equipped with safety valve	127	
Doors	Not as specified (stainless steel)	128	
	Not provided with hinges	129	
	Magnetic latches missing	130	
	Does not provide access to boiler	131	
	Handles missing	132	
Controls	Missing or not as specified	133	
	Controls not accessible for servicing	134	
	Piping and valves not concealed in cabinet base	135	
	Valve working pressure not cast on side		208
Pressure gage	Missing	136	
	Dial not graduated as specified	137	
	Not visible to operator	138	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Data-name plate	Missing or not as specified		209
	Information incomplete or not legible		210
	Not located so as to be readily visible to operator		211
User instructions	Missing, incomplete, not legible, not specified type or size, not affixed in the prescribed manner	139	

4.2.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in table I. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of tilting kettle assemblies. The sample unit shall be one tilting kettle assembly. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 End item testing.

4.2.4.1 Initial production testing. The initial production end item shall be tested as specified in 4.4.3 and 4.4.4. Any failure of the initial production end item shall be cause for rejection of the end item, necessary design and construction changes, and repetition of testing of the next initial production end item until both tests are sucessfully passed.

4.2.4.2 Operational and hydrostatic testing. Every end item shall be tested as specified in 4.4.1 and 4.4.2. Any failure of any end item to pass either test shall be cause for rejection of that end item.

4.2.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: preservatives missing or not dried, surface not cleaned, openings not sealed, crate not anchored, blocked and braced as specified
Content	Number of intermediate containers is more or less than required

4.3 Code and standard compliance. Certifications, certified test reports, or listing marks for standards submitted in accordance with 3.2 shall be examined and validated as proof of compliance. Prior to Government approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the convection ovens he proposes to furnish under this document, meet the applicable requirements of UL 197; ANSI 83.15; sections IV and VIII of the ASME Boiler and Pressure Vessel Code; and NSF Standard No. 4.

4.3.1 UL. Acceptable evidence of meeting the applicable requirements of UL 197 shall be the UL label, the UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the kettle assembly offered has been tested and conforms to the applicable requirements of UL 197.

4.3.2 ANSI. Acceptable evidence of meeting the applicable requirements of ANSI Z83.15 may be a photostatic copy of the AGA Appliance Certificates; a listing of the tilting kettle assembly in the AGA Directory of Certified Appliances and Accessories; or a certified test report from a recognized, independent testing laboratory acceptable to the Government stating that the tilting kettle assembly has been tested and conforms to ANSI Z21.46.

4.3.3 ASME code. Acceptable evidence of meeting the applicable requirements of the ASME Boiler and Pressure Vessel Code, Section IV and VIII shall be the ASME symbol stamped on the boiler and kettles, respectively.

4.3.4 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished tilting kettle assembly.

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b. A certification for the tilting kettle assembly issued by NSF under their special one-time contract evaluation/certification service.

c. A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the tilting kettle assembly is being procured indicating that the tilting kettle assembly has been tested and conforms to NSF Standard No. 4.

4.4 Methods of inspection.

4.4.1 Operational test. The tilting kettle assembly shall be given an operational test for not less than 15 minutes to determine that controls and valves operate as specified in 3.4.2.3 and 3.4.3.2, that there is no evidence of leakage as specified in 3.4, and that the safety valve functions are in accordance with 3.4.2.1.1. During this test, the temperature shall be checked for conformance with 3.4 and the Btu input of the steam generating boiler shall be checked for conformance with 3.4.3. Pressure gage accuracy shall be tested at 30 psig to determine conformance with 3.4.2.7. Automatic drain unit shall be tested to function in accordance with 3.4.2.9. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.2 Hydrostatic test. Each tilting kettle and steam generator boiler shall be tested in accordance with the applicable section of the ASME Boiler and Pressure Code at the hydrostatic pressure specified herein. Any unit that fails to pass this test shall be rejected. Certification in accordance with 4.3.3.3 will be acceptable evidence that the unit has satisfactorily passed this test.

4.4.3 Capacity test. Each tilting kettle shall be tested to determine compliance with capacity requirements of 3.4. The test shall be performed by filling the kettle with water from a container of known capacity. Any noncompliance with specified requirements shall constitute failure of the test.

4.4.4 Deformation test. The tilting kettle shall be loaded with weights equivalent to 1-1/2 times the weight of its water capacity for 15 minutes and the trunnion support(s) shall be checked to determine conformance with the deformation resistance requirements of 3.4.1.4. Any nonconformance with the requirements specified shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. All exposed, uncoated, ferrous metal surfaces except corrosion-resisting steel of each cooker shall be cleaned using process C-1 of MIL-P-116 and thoroughly dried.

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5.1.1.2 Application of preservation. Surfaces cleaned as specified shall be coated with type P-14 preservative of MIL-P-116.

5.1.1.3 Sealing of openings. All openings, such as inlets, outlets, vents, valves, and the face of the steam gage shall be sealed with tape conforming to type III, class 1, or type IV of PPP-T-60.

5.1.1.4 Doors. The doors of the cabinet base shall be securely closed so as to prevent their opening while in transit.

5.1.2 Commercial. Each tilting kettle assembled shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Each tilting kettle assembly, preserved as specified in 5.1, shall be packed in a plywood crate conforming to type I, class 2, style a of MIL-C-104. Anchoring, blocking, and bracing shall be in accordance with the appendix of MIL-C-104.

5.2.2 Level B packing. Each tilting kettle assembly, preserved as specified in 5.1, shall be packed in a cleated plywood shipping container conforming to domestic type, style A or B of PPP-B-601. The tilting kettle assembly shall be secured with lumber blocking and bracing and a minimum space of 1 inch shall be maintained between the legs of the tilting kettle assembly and the inside surface of the bottom of the shipping container.

5.2.3 Commercial packing. Kettles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.4 Marking. In additin to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The tilting kettle assembly is intended for cooking of food in kitchens of enlisted personnel dining facilities.

6.2 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. When autommatic drain unit is required (see 3.4.2.9).
c. Electrical requirements, if other than specified (see 3.4.4).
d. Government approved manual number to be included on data-name plate (see 3.6.1).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

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6.3 Subject term (key word) listing.

Food service equipment
Kettles
Kitchen equipment
Steam cooker
Steam kettle

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD, TS
Air Force - 82, 84
DLA - GS

Preparing activity:

Army - GL
Project No. 7310-0741

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-K-43943A	Kettles, Steam Jacketed; Twin; 5-Gallon, Tilting, Cabinet Mtd.

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, City, State, ZIP Code)

4. TYPE OF ORGANIZATION (Mark one)

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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